ABSTRACT

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The Philippines experiences frequent tropical cyclones throughout the year, regularly testing the effectiveness of local disaster preparedness and evacuation systems. In Kawit, Cavite, the nine coastal barangays are particularly vulnerable to typhoon-related flooding due to their coastal location and dense residential distribution. However, limited network-based studies and outdated evacuation planning hinder the optimization of evacuation routes and facility allocation in the area. This study analyzed the evacuation planning in Kawit's coastal barangays by examining road network connectivity, evacuation center accessibility, and facility capacity relative to the estimated evacuee population. A graph-theoretical approach was applied using QGIS. Dijkstra's Algorithm was used to determine the optimal evacuation routes from building cluster centroids to evacuation centers, the Betweenness Centrality identified critical nodes that function as key connectors and potential bottlenecks in the evacuation network, and the Crowd Density Ratio was computed to assess whether evacuation centers meet the Humanitarian Standard for space allocation. The results revealed that seven barangays exhibited uneven spatial distribution of residential areas and evacuation centers, resulting in varying levels of accessibility. Three barangays showed well-integrated evacuation centers within the road network, while several barangays relied heavily on critical intersections and major roads that may become bottlenecks during evacuation. Furthermore, only two barangays met the recommended humanitarian space standard, while the majority were at risk of overcrowding. The findings highlight the need to optimize evacuation routes and planning through improved network connectivity, strategic placement of evacuation centers, and enhancement of facility capacity to ensure safer and more efficient evacuation systems.